

Equivalence b/w two FSM's

The 2 finite automata are said to be equivalent if both the automata accept the same set of strings over an I/P set Z .

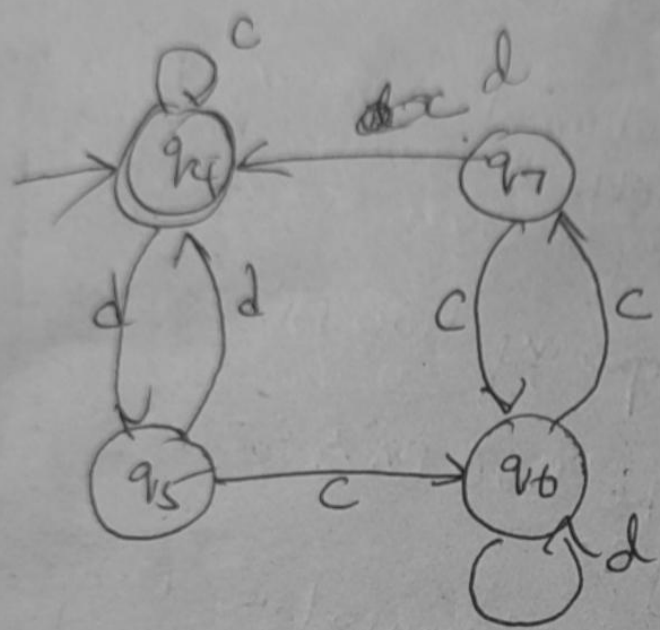
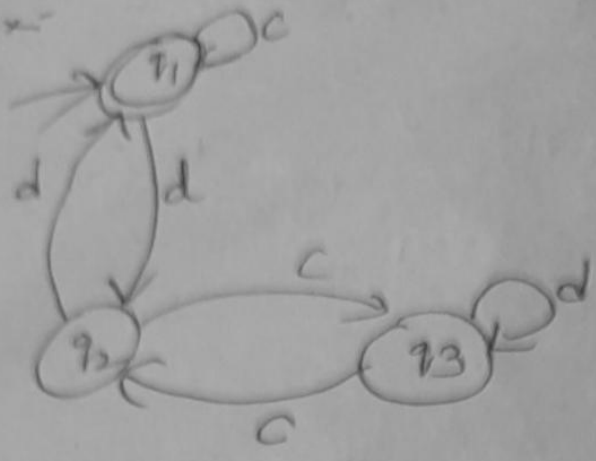
Method for Comparing two FA's

Let M & M' be two FA's & Z is a set of I/P strings

- 1) Construct a transition table with pairwise entries where (q, q') where $q \in M$, & $q' \in M'$ for each I/P symbol
- 2) If ~~the~~ one of the states in the pair is final & the other is non-final then terminate the construction of transition table as ~~they~~ ^{FA's} are not equivalent.
else repeat
- 3) ~~repeat~~ ^{repeat} until the construction until no new pair appears.

check whether 2 FAs are equivalent or not?

ex-



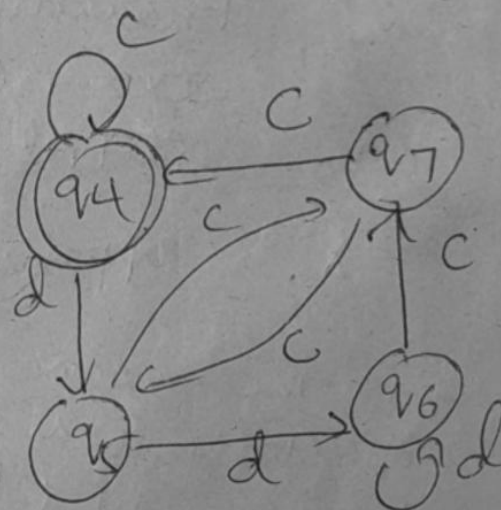
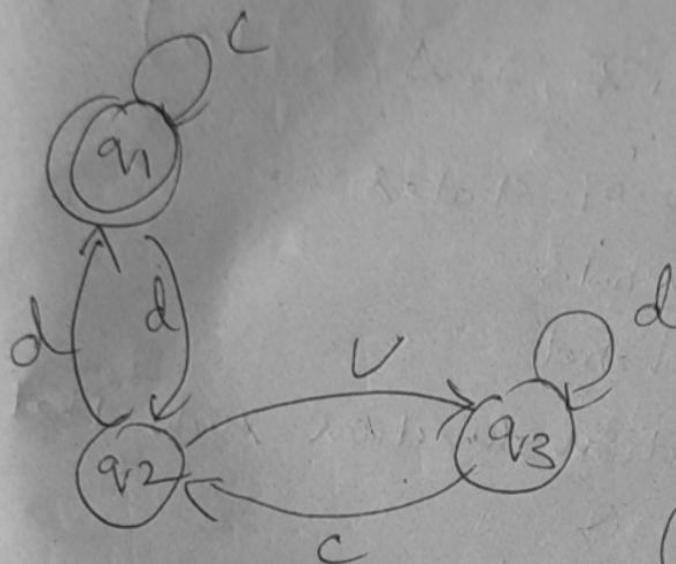
	c	d
q1 q4	q1 q4 fs fs	q2 q5 fs fs IS IS
q2 q5	q3 q6 IS IS	q1 q4 fs fs
q3 q6	q2 q7 IS IS	q3 q6 IS IS
q2 q7	q3 q6 IS IS	q1 q4 fs fs

q_1 is final state in M & q_6 is not a final state in M' .

Hence, stop constructing the transition table

$\therefore M$ & M' are ~~not~~ equivalent

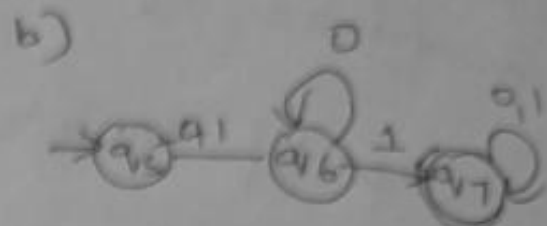
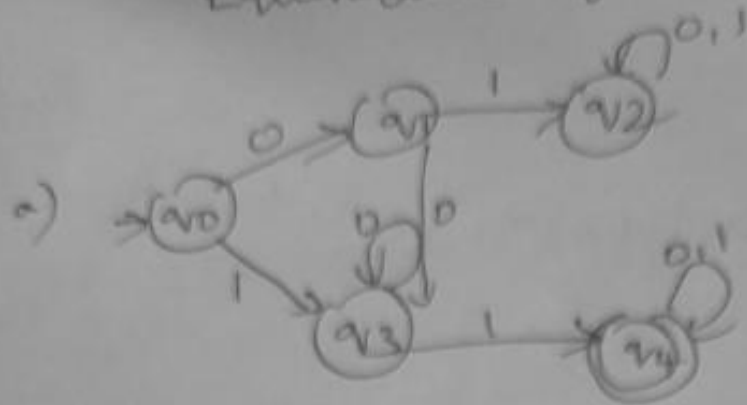
to check whether
Q) following FA's are equivalent or not?



	c	d
$q_1 q_4$	$q_1 q_4$ FS FS	$q_2 q_5$ IS IS
$q_2 q_5$	$q_3 q_7$ IS IS	$q_1 q_6$ FS IS

$\therefore$ given FA's are not equivalent

Equivalence HW 2 FSM's



	0	1
$q_0 q_5$	$q_1 q_6$	$q_3 q_6$
$q_1 q_6$	$q_3 q_6$	$q_2 q_7$
$q_3 q_6$	$q_3 q_6$	$q_4 q_7$
$q_2 q_7$	$q_2 q_7$	$q_2 q_7$
$q_4 q_7$	$q_4 q_7$	$q_4 q_7$

equivalent